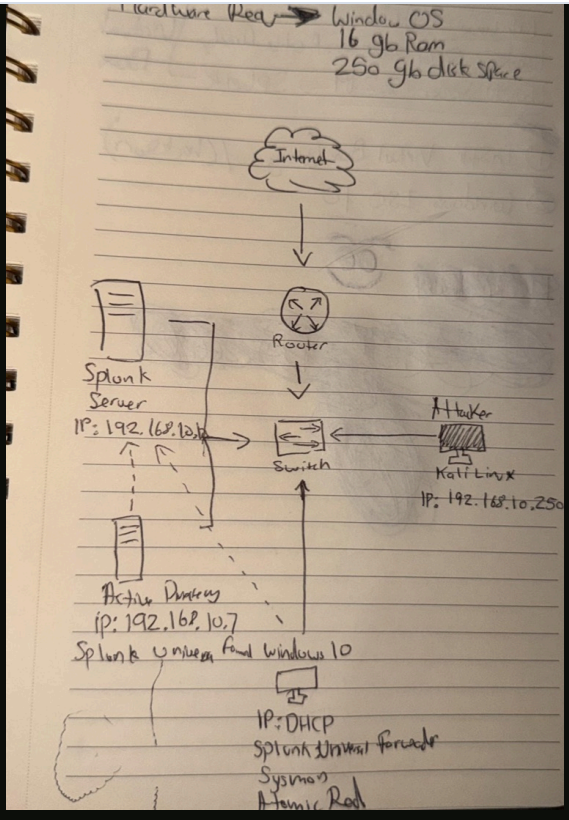
TECHNICAL PROJECT REPORT

Active Directory Enterprise Lab: Build, Detection & Troubleshooting

Designing, deploying, and defending a Windows domain with Splunk-based monitoring, validated through controlled adversary emulation and a full troubleshooting runbook.



Network Structure Framework — the lab designed before deployment.

Prepared by	Sukhwinder Singh	**Document type**	Portfolio / Technical Report
Focus area	IT Administration · Blue Team	**Environment**	4 VMs · 192.168.10.0/24
Platform	VirtualBox · Win Server 2022 · Splunk · Kali	**Version**	1.0

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DOCUMENT SUMMARY

This report documents the end-to-end construction of a functional Active Directory environment and the operational practices that surround it. It covers the network design, the deployment of a Splunk SIEM with endpoint telemetry, the build-out of the domain, a controlled attack used to *validate* that the monitoring works, the hardening applied afterward, and a runbook of the real-world incidents an administrator resolves day to day. Figures throughout are screenshots captured directly from the live lab.

1. Executive Summary

This project delivers a complete, enterprise-style Active Directory (AD) environment built from the ground up on four virtual machines, instrumented with a Splunk security-monitoring pipeline, and exercised through a controlled attack to prove the monitoring actually detects malicious activity.

The work was organized into six phases: network design, monitoring deployment, Active Directory build, detection validation, security hardening, and troubleshooting. Rather than treating offensive techniques as the goal, the attack simulation in Phase 4 serves a defensive purpose — it confirms that the Splunk Universal Forwarder and Sysmon are capturing the right telemetry, and it surfaces the visibility gaps that remain. The findings then drive the hardening in Phase 5.

Key outcomes:

- A four-host lab on a single `192.168.10.0/24` network: a Splunk SIEM, a Windows Server 2022 domain controller, a Windows 10 client, and a Kali attacker.
- End-to-end log forwarding verified, with Sysmon providing detailed endpoint telemetry to Splunk over port `9997`.
- A brute-force attack over RDP was detected cleanly in Splunk as **20 failed-logon events (4625)** followed by a single successful logon (4624), with the source traced back to the attacker.
- Adversary emulation with Atomic Red Team, mapped to the MITRE ATT&CK framework, identified a concrete **visibility gap** for local-account creation — a prioritized target for future detection.
- Hardening applied least-privilege group management, logon-hour restrictions, and automatic account expiration.
- A documented runbook resolves four common production incidents, including recurring account lockouts and a broken workstation trust relationship.

2. Objectives & Scope

Objectives

- Design and document a realistic network before building it.
- Deploy and configure Active Directory Domain Services, DNS, and Group Policy on Windows Server 2022.
- Stand up a Splunk SIEM and forward Windows and Sysmon telemetry from every endpoint.
- Validate the detection pipeline against a real attack and quantify what is — and is not — visible.
- Apply security hardening informed by those results.
- Produce a repeatable troubleshooting runbook for common domain incidents.

In scope

Virtualized infrastructure, domain services, endpoint logging, SIEM ingestion, a single controlled attack path (RDP brute force) and adversary emulation (Atomic Red Team), account and permission administration, and incident resolution.

Out of scope

Production-grade high availability, multi-site/multi-domain forests, external perimeter security, and any activity outside this isolated lab network. All testing was performed against systems built and owned for this project.

3. Lab Architecture & Environment

All machines run under Oracle VirtualBox on a single NAT network, emulating a flat corporate LAN. The Splunk server and domain controller use static addresses; the client is assigned a static address for predictable forwarding and testing. The Kali host sits logically outside the domain's trust as the attacker.

Table 1. Host inventory and roles.

Host	Operating system	IP address	Role & services
Splunk	Ubuntu Server	`192.168.10.10`	SIEM indexer; receiver on `:9997`, web console on `:8000`
ADDC01	Windows Server 2022	`192.168.10.7`	Domain Controller, DNS, Group Policy; Universal Forwarder + Sysmon
target-PC	Windows 10	`192.168.10.100`	Domain-joined workstation; Universal Forwarder + Sysmon
kali	Kali Linux	`192.168.10.250`	Attacker — Crowbar (brute force) and Atomic Red Team

DATA FLOWS

- **Telemetry:** the domain controller and client forward Windows Event Logs and Sysmon events to the Splunk indexer on `:9997`.
- **Authentication / name resolution:** the client resolves names and authenticates against the domain controller (DNS pointed at `192.168.10.7`).
- **Attack path:** Kali targets the client's exposed RDP service (`:3389`).

4. Phase 1 — Design & Network Configuration

A build is only as sound as its plan. The first phase defined the entire environment on paper, then implemented the networking so the four machines behaved like hosts on one corporate subnet.

1.1 Network design

The **Network Structure Framework** (cover figure) mapped every host, its address, and the traffic between them before any VM was created. Designing first forces a clear understanding of the environment and how it would be secured — and mirrors the diagramming an administrator is expected to produce on demand.

1.2 Provisioning & integrity

Each installer was validated with a SHA-256 hash against its published value before use, ensuring only genuine, untampered media entered the lab. All four virtual machines were then provisioned in VirtualBox and attached to a common NAT network, enabling snapshots for safe, reversible testing.

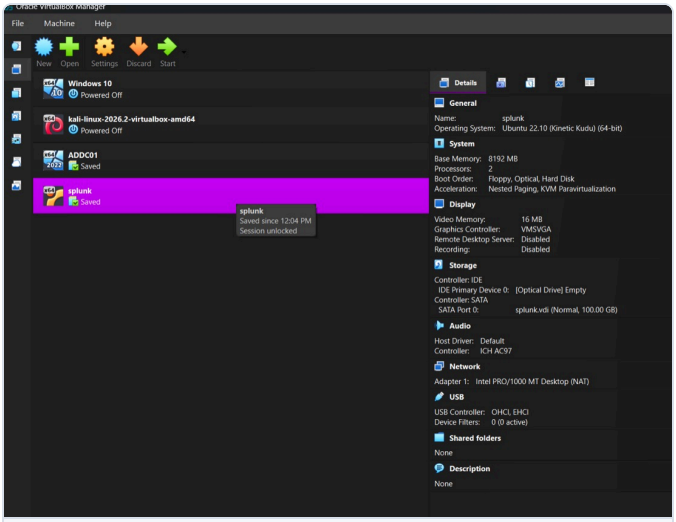


Figure 1. The four-machine fleet in VirtualBox — client, Kali, domain controller, and Splunk server.

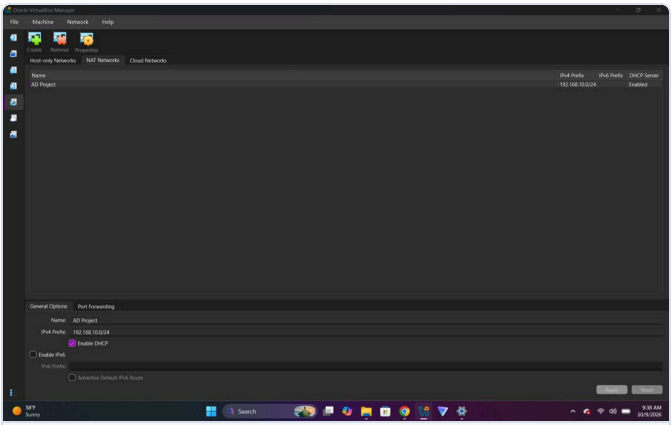


Figure 2. The NAT network “AD Project” on 192.168.10.0/24 — the shared subnet for all hosts.

1.3 Static addressing

On the Ubuntu/Splunk server, a static IP was configured by editing the Netplan file `00-installer-config.yaml` and applying it; connectivity was confirmed with `ip a` and a successful ping to the internet. A dedicated service user and a mounted shared folder were created to move installation media onto the server cleanly.

5. Phase 2 — Monitoring Pipeline: Splunk & Sysmon

With the network in place, the next phase built the security-monitoring backbone: a Splunk SIEM ingesting detailed endpoint telemetry from every Windows host.

2.1 Splunk SIEM

Splunk Enterprise was installed under `/opt/splunk` on the Ubuntu server to act as the indexer and search platform. Receiving was enabled so that forwarders could deliver events on port `9997`.

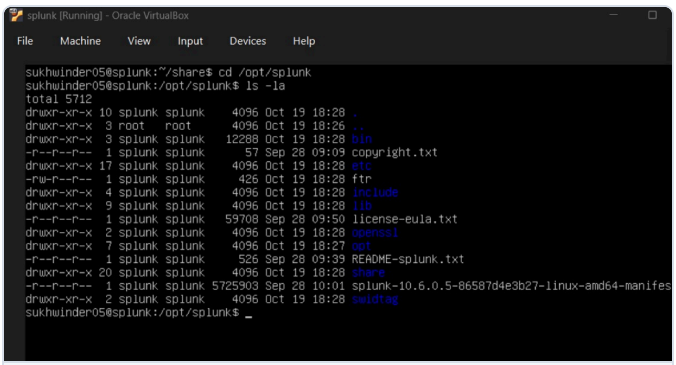


Figure 3. Splunk Enterprise installed on the Ubuntu server.

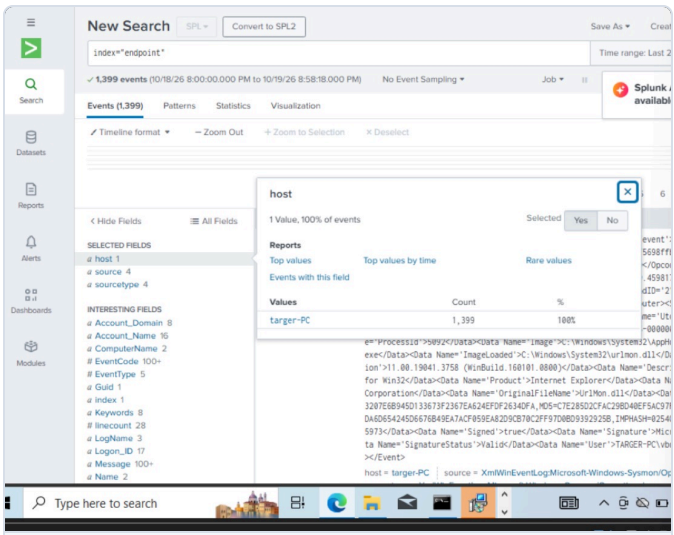


Figure 4. Indexes receiving forwarded data on port 9997, confirming the pipeline is live.

2.2 Endpoint telemetry — Universal Forwarder & Sysmon

The Splunk Universal Forwarder and **Sysmon** were installed on both the Windows 10 client and the Windows Server 2022 domain controller. Sysmon enriches the standard Windows logs with high-fidelity detail — process creation, command lines, hashes, and network connections — which is the raw material for meaningful detections. Forwarding was controlled through a *local* copy of `inputs.conf`; the default configuration was deliberately left untouched to avoid breaking the application, a supportable practice in production environments.

2.3 Service permissions

The forwarder initially ran as the low-privilege `NT` service account, which lacked the rights to read every required log. The service logon identity was changed so the full event set could be collected — a practical lesson in least privilege versus operational need.

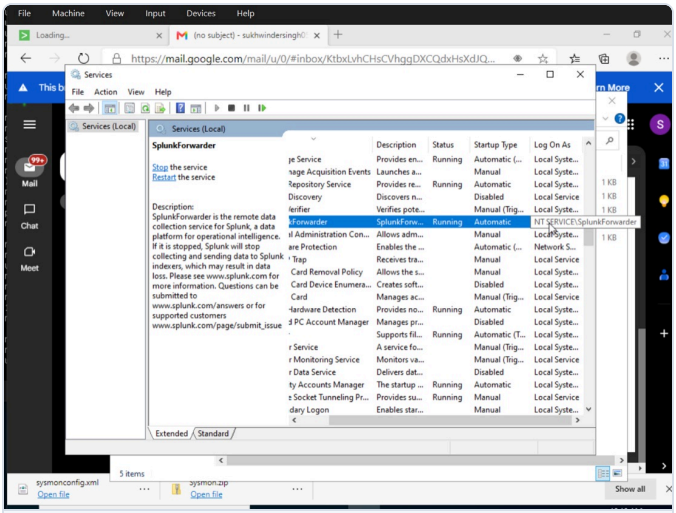


Figure 5. Sysmon and the Splunk forwarder running after the service-account fix.

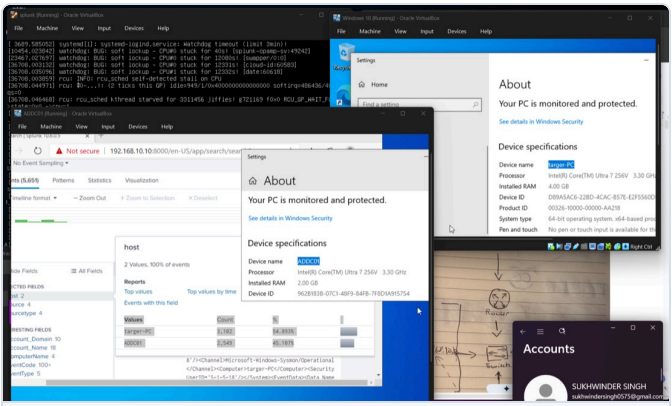


Figure 6. Sysmon fully deployed on both the client and the domain controller.

6. Phase 3 — Active Directory Deployment

This phase stood up the domain the way a real organization runs it: a promoted controller, a departmental structure, managed users, integrated DNS, and a joined client.

3.1 Promoting the domain controller

Active Directory Domain Services and role-based access control services were installed on the Windows Server 2022 host, which was then promoted to a domain controller with a new `.local` root domain. Signing in with the domain administrator account confirmed a successful promotion.

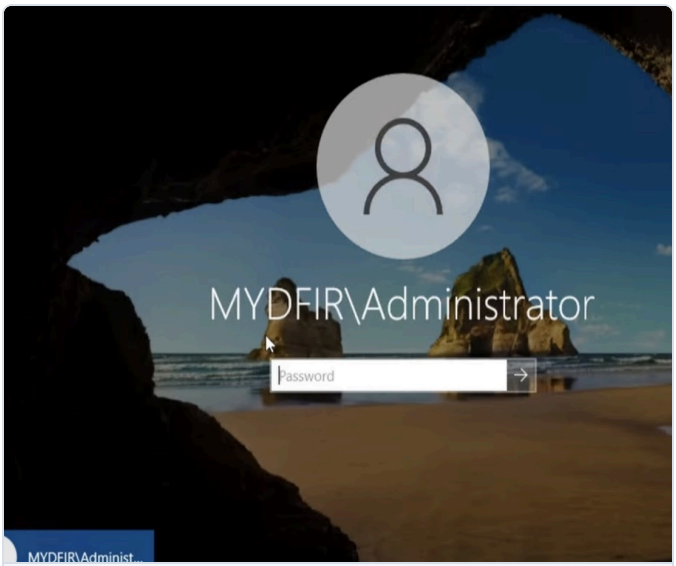


Figure 7. Domain administrator sign-in confirms the controller is promoted.

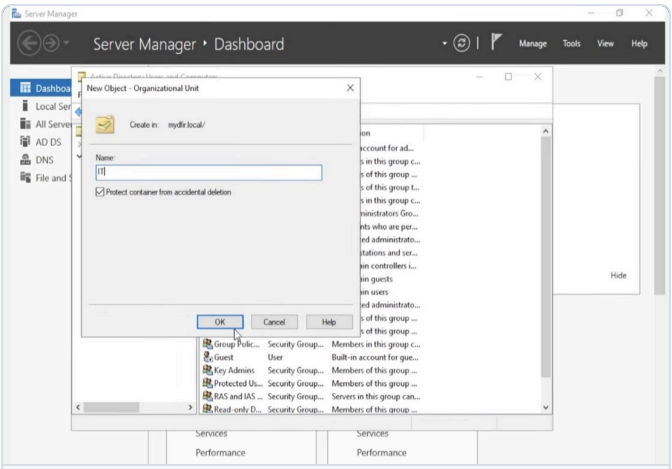


Figure 8. Creating Organizational Units to separate IT, HR, and other departments.

3.2 Users, groups & DNS

Organizational Units were created to model the company's departments, and user accounts were placed into the correct OUs (for example, a new user added to IT). DNS was then changed from a public resolver (`8.8.8.8`) to the Active Directory domain service, so name resolution runs through the controller — a prerequisite for domain membership.

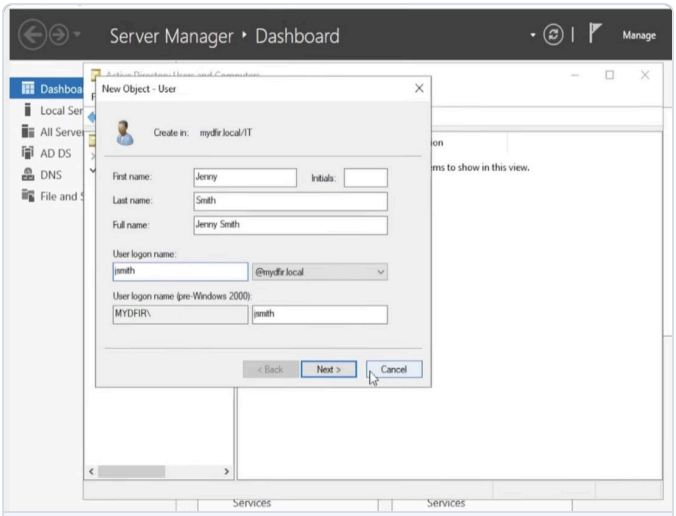


Figure 9. Creating a domain user and assigning the correct department OU.

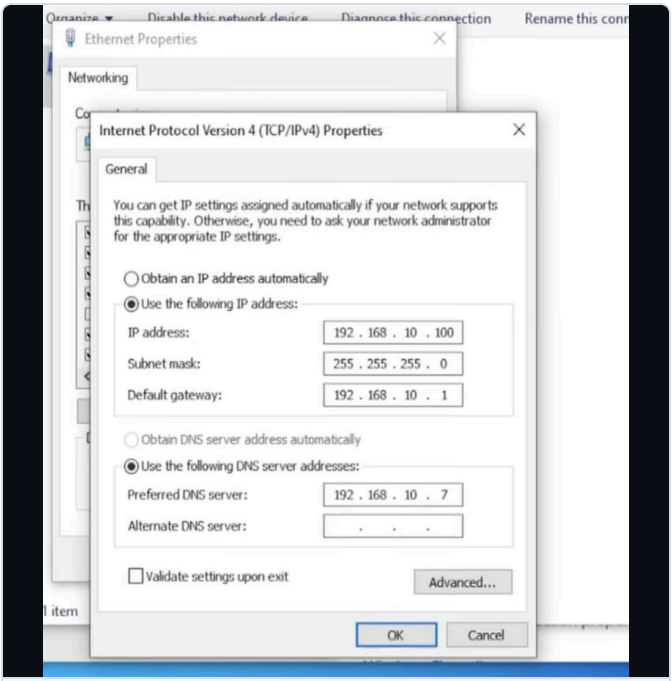


Figure 10. Pointing DNS at the Active Directory domain service.

3.3 Joining the client

With DNS configured, the Windows 10 client was joined to the domain and every AD user could authenticate on it. The client's `ipconfig` confirmed a static address with DNS pointed at the controller.

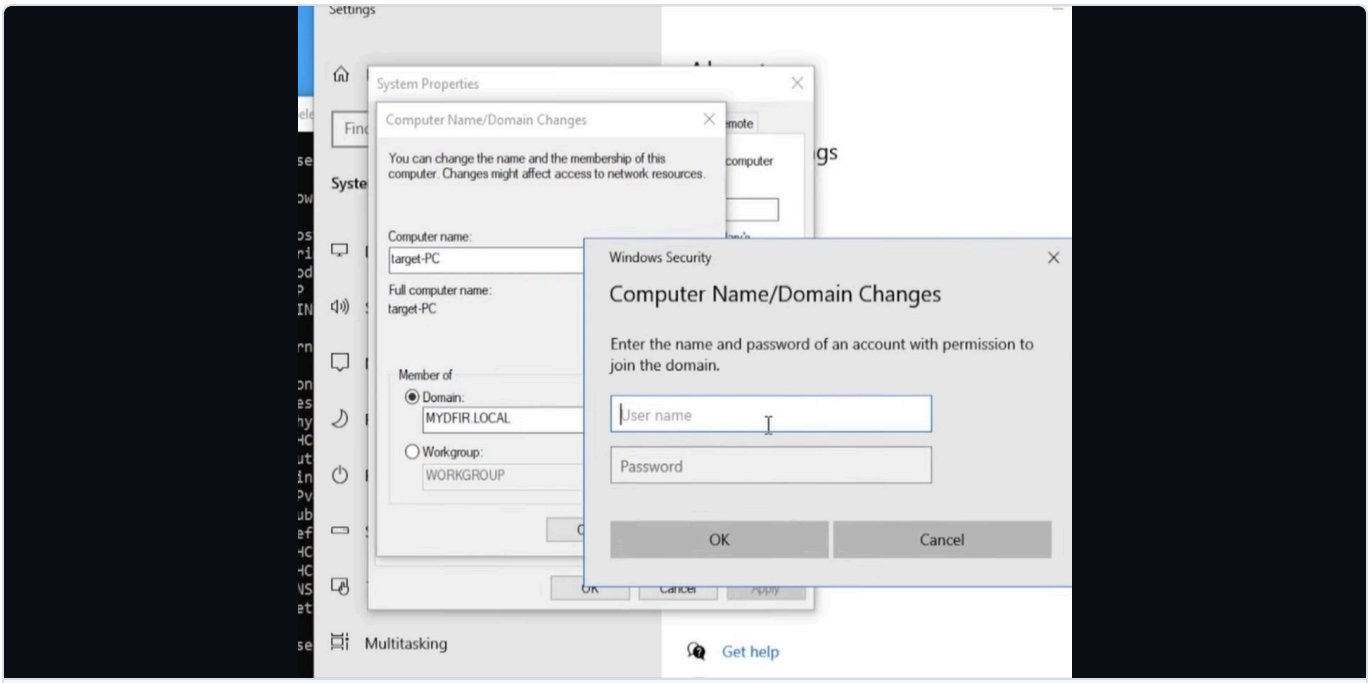


Figure 11. Joining the Windows 10 workstation to the domain — a standalone PC becomes a managed member.

7. Phase 4 — Attack Simulation & Detection Validation

A monitoring pipeline is only trustworthy once it has caught something real. This phase ran a controlled attack from Kali, verified detection in Splunk, and then used adversary emulation to measure the remaining blind spots.

4.1 Brute-force attack (RDP)

Remote Desktop was enabled for selected users on the client. From Kali, the **Crowbar** tool drove an RDP brute force against account `tsmith` at `192.168.10.100` using a prepared password wordlist. The tool worked through the list and returned `RDP-SUCCESS`, recovering a valid credential.

```
└─$ crowbar -b rdp -u tsmith -C passwords.txt -s 192.168.10.100/32
Trying 192.168.10.100:3389
RDP-SUCCESS : 192.168.10.100:3389 - tsmith:P@ssw0rd1!
```

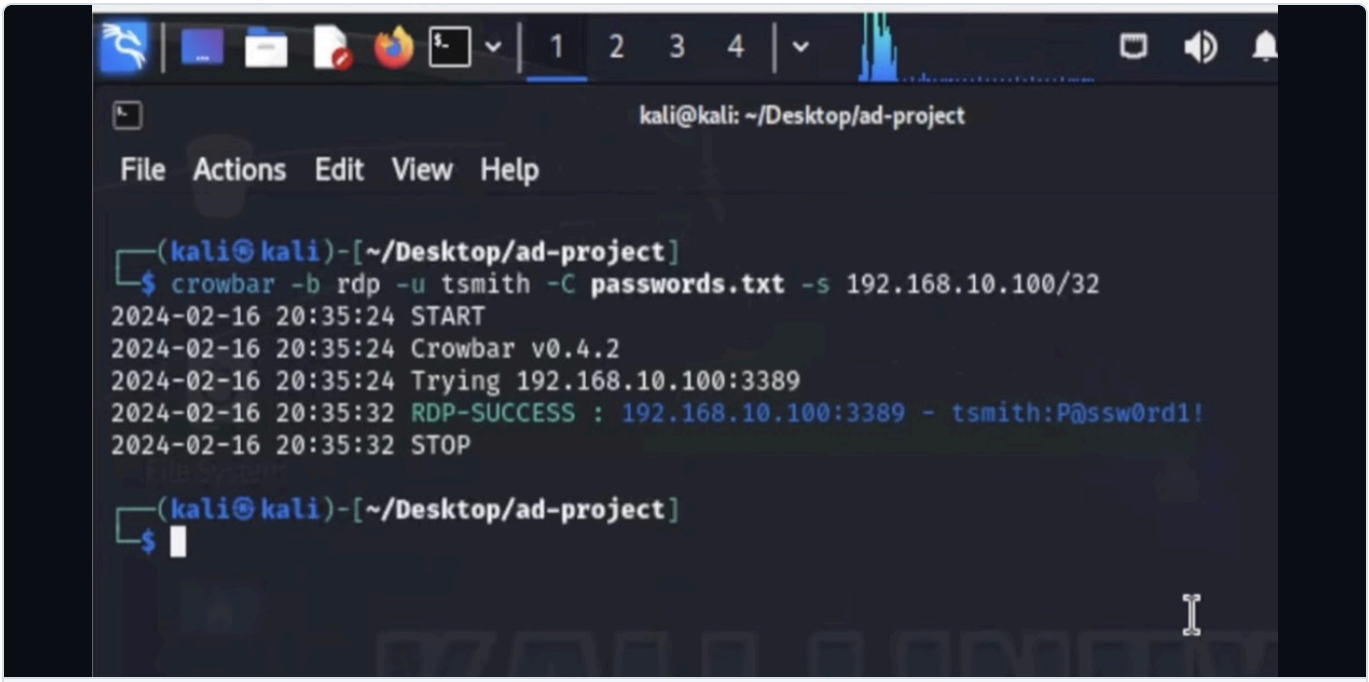


Figure 12. Crowbar returns RDP-SUCCESS against the Windows client.

4.2 Detection in Splunk

In Splunk, the attack produced a clear, unambiguous signature. Twenty **Event 4625** (failed logon) records appeared, followed by a single **Event 4624** (successful logon). Critically, all failures occurred in the same instant — a pattern characteristic of automated attacks rather than human error — and the successful logon event identified the **source address of the Kali attacker**.

Table 2. Windows security events generated by the brute force, as observed in Splunk.

Event ID	Count	Meaning	Significance
4625	20	Failed logon	Burst at one instant → automated brute force
4624	1	Successful logon	The breach; source traced to attacker IP
4634	1	Logoff	Session teardown
4776	1	Credential validation	Supporting authentication evidence

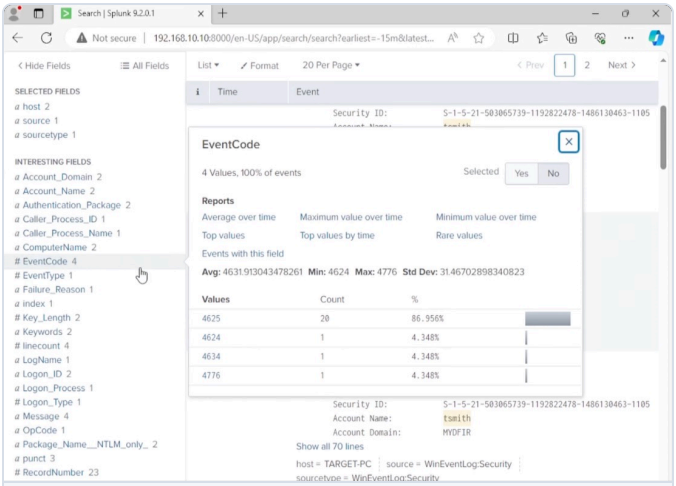


Figure 13. Splunk shows 20 × Event 4625 (≈87% of events) for tsmith on the target.

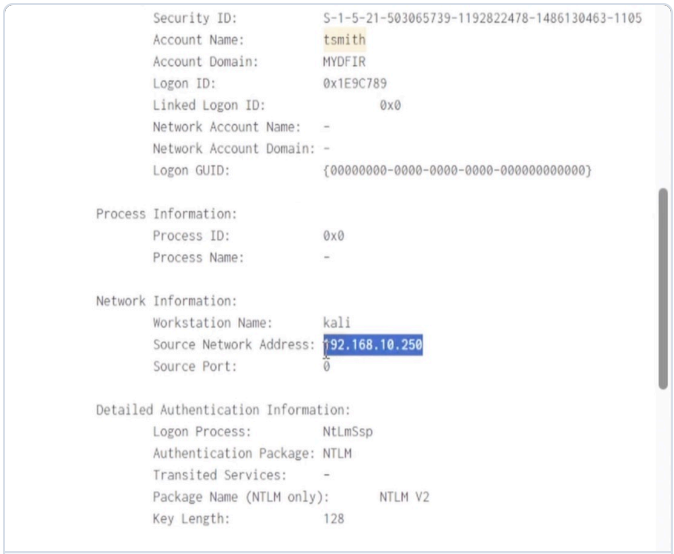


Figure 14. The successful 4624 event identifies the source — the Kali attacker's address.

4.3 Adversary emulation — Atomic Red Team & MITRE ATT&CK

To measure coverage beyond a single attack, **Atomic Red Team** was installed (after adding a scoped Microsoft Defender exclusion so the test files were not quarantined). Its "atomics" are small, safe tests, each tied to a technique ID in the MITRE ATT&CK framework.

T1110 — Brute Force (RDP) | T1136.001 — Create Account: Local | T1059.001 — PowerShell | T1059.003 — Windows Command Shell

The **Create Account (T1136.001)** test created a local account, but the activity did *not* appear in Splunk. This negative result was the most valuable finding of the phase.

> **Finding — visibility gap (local account creation)**
>
> Under the current configuration, creation of a local account produces no visible telemetry in Splunk. An attacker establishing persistence this way would go undetected. This is a prioritized target for a new detection rule, not a dead end — it is precisely the kind of blind spot adversary emulation is designed to reveal.

By contrast, the **PowerShell / command-shell techniques (T1059)** were captured in detail by Sysmon — the full command line was visible in Splunk — meaning alerts can be built on that activity immediately.

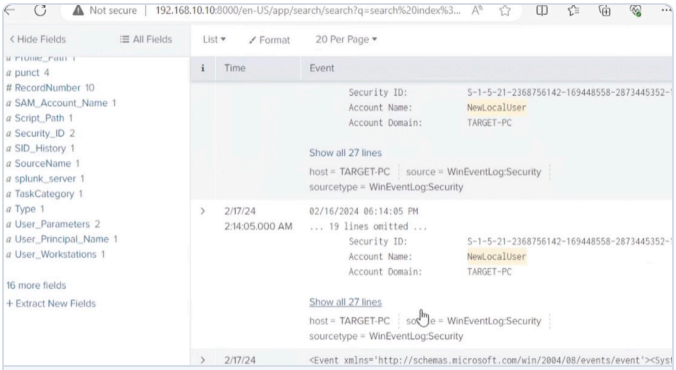


Figure 15. The local-account test produced no Splunk results — a documented visibility gap.

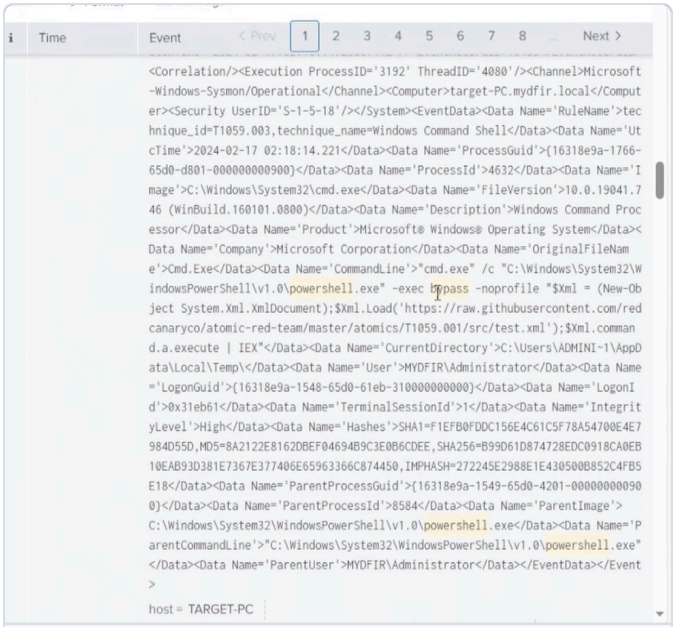


Figure 16. Sysmon captures the PowerShell technique in full — a ready basis for alerting.

8. Phase 5 — Security Hardening

Informed by the validation results, the environment was hardened using native Active Directory and Group Policy controls that reduce standing access and limit the window of risk.

5.1 Group-based access (least privilege / RBAC)

Permissions were assigned to **security groups** rather than to individual users, so access can be granted and revoked for many people at once and remains auditable at scale. A realistic set of simulated users and data was created to represent a running organization.

5.2 Logon-hour restrictions

Time-based access control was applied so that users cannot sign in during a defined maintenance window, protecting backups and software updates from interference.

5.3 Automatic account expiration

Contractor accounts were configured to **expire on their end date**, so access terminates automatically when a contract ends — removing the risk of forgotten, lingering accounts.

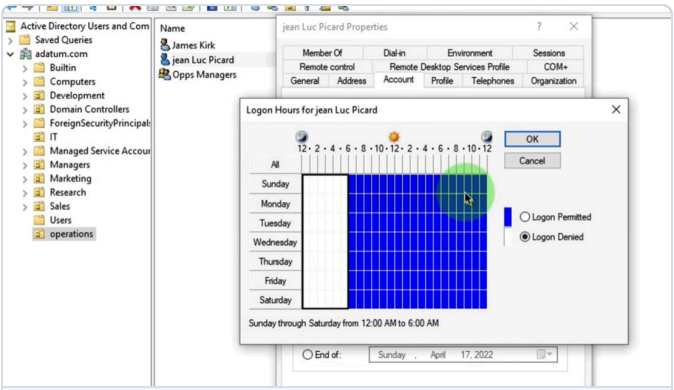


Figure 17. Logon-hour restrictions limiting when an account may sign in.

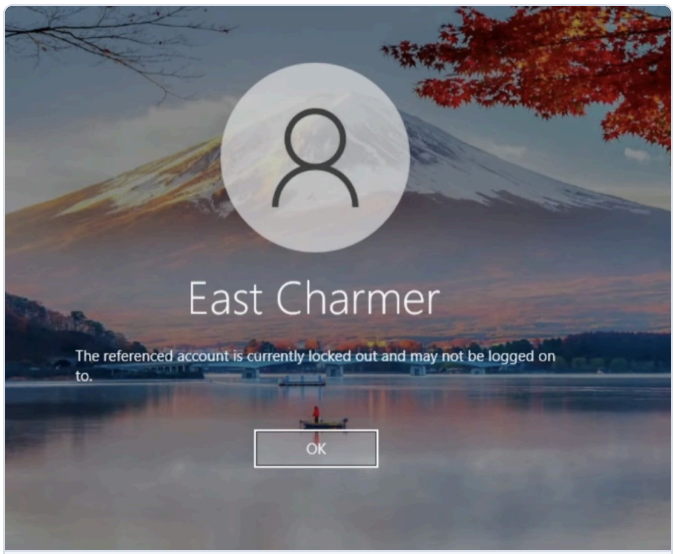


Figure 18. Account expiration ending contractor access automatically on the due date.

9. Phase 6 — Troubleshooting Runbook

Operating a domain means resolving the issues users actually encounter. Each incident below is documented as symptom → diagnosis → resolution, including escalation paths when the obvious fix does not hold.

ACCOUNT Incident 1 — Recurring account lockout

SYMPTOM A user cannot sign in; the account is locked. After a password reset and unlock, it locks again within minutes.

DIAGNOSIS A recurring lockout is not a password problem. **Event Viewer** security logs are reviewed to identify which computer is generating the failed logons — typically a stale cached credential, a mapped drive, or a service still using the old password.

RESOLUTION Correct the *source* of the bad credentials rather than repeatedly unlocking the account. Once the source is fixed, the unlock holds.

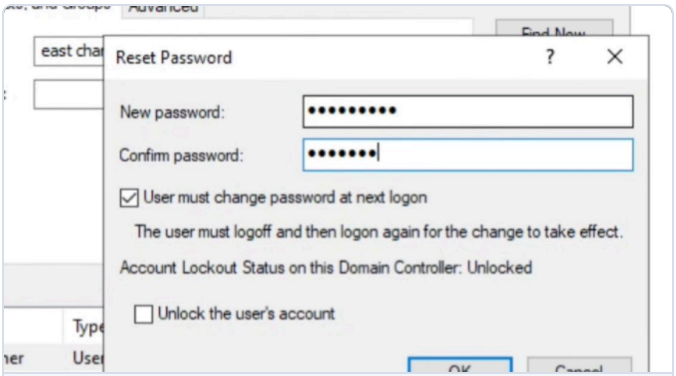


Figure 19. The account shown locked out in Active Directory.

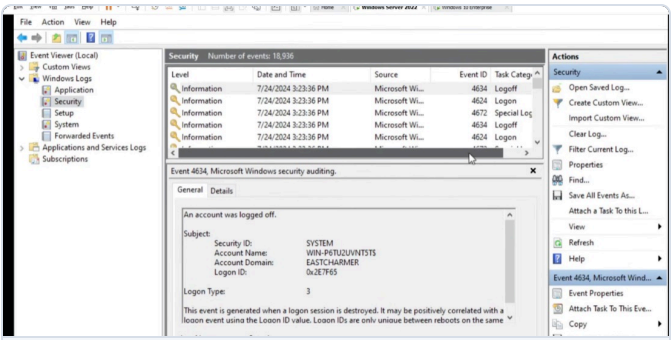


Figure 20. Event Viewer security logs used to trace the lockout source.

ACCOUNT Incident 2 — Expired account

SYMPTOM A valid user suddenly cannot log in, with no lockout and a correct password.

DIAGNOSIS The account's **"Account expires"** date has passed — common for contractors, and the expected result of the hardening applied in Phase 5.

RESOLUTION If the person is continuing, extend or remove the expiration; if not, confirm the cutoff was intentional. Access then matches the contract.

ACCESS Incident 3 — Access denied to a shared resource

SYMPTOM A new HR hire receives **"access denied"** when creating files in the HR share.

DIAGNOSIS The user's **Member Of** tab is empty. Comparison with a working HR peer shows the peer belongs to the `$HR-Staff` group, and the folder's NTFS security grants that group full control. The new hire was never added.

RESOLUTION Add the user to `$HR-Staff` so access is inherited through the group, then verify against the folder's NTFS permissions — clean, auditable, group-based access.

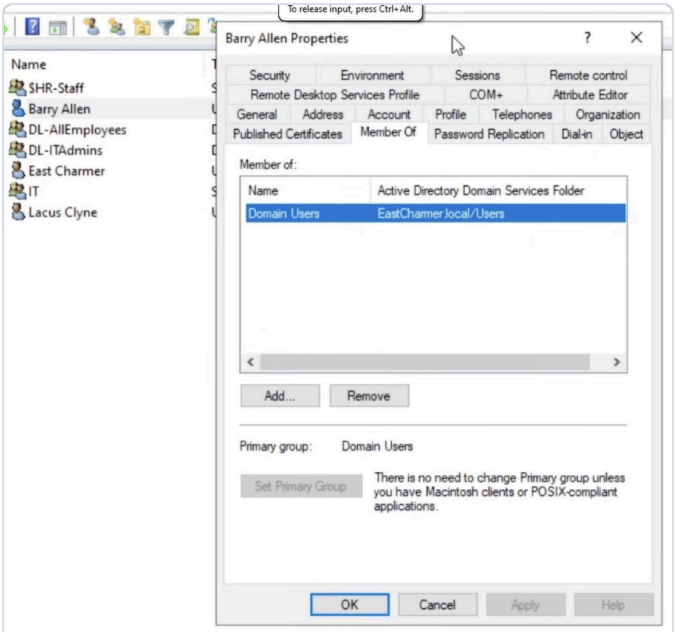


Figure 21. Comparing the new hire's group memberships against a working HR peer.

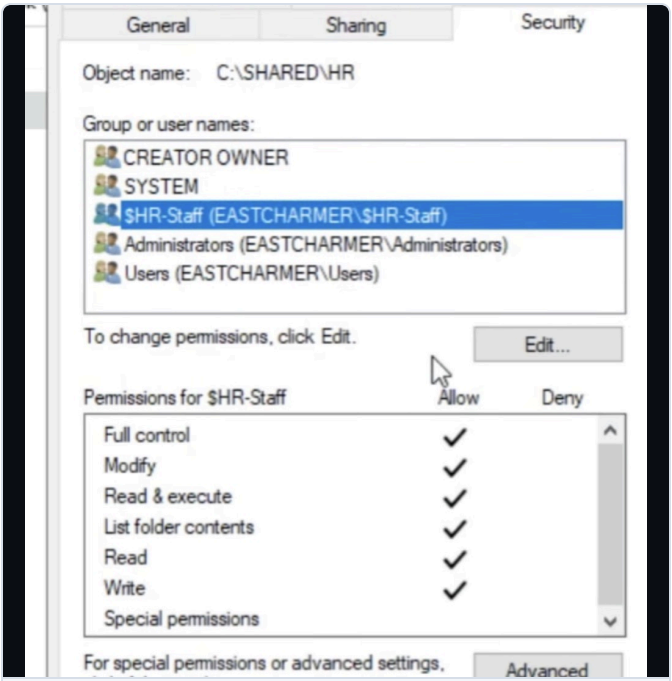


Figure 22. The HR folder's NTFS security: the $HR-Staff group has full control.

HOST Incident 4 — Broken workstation trust relationship

SYMPTOM At sign-in: *"The security database on the server does not have a computer account for this workstation trust relationship."* Domain credentials are rejected.

DIAGNOSIS The computer's secure channel with the domain has broken, typically after being disconnected for an extended period; the machine's account secret no longer matches the domain's.

RESOLUTION (1) Sign in with the **local** admin account (`.\admin`). (2) Move the PC to a workgroup and reboot. (3) **Delete the stale computer object in AD** — if the same name is reused, the old object must be removed first or the rejoin conflicts. (4) Rejoin the domain and sign in.

Operational note

Knowing that the stale computer object must be deleted in AD *before* rejoining with the same name is the difference between a five-minute fix and hours of conflicts.

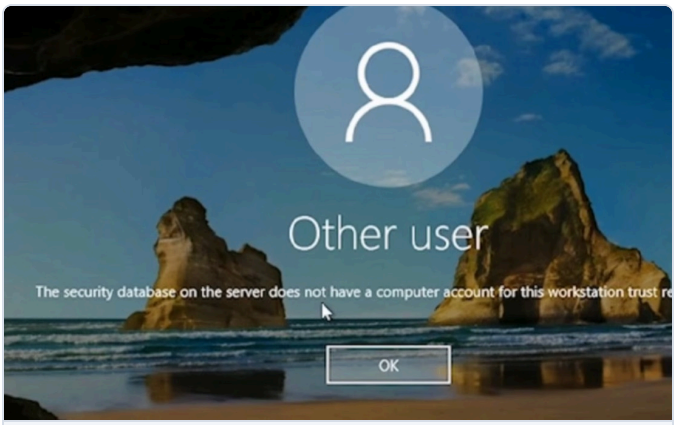


Figure 23. The workstation trust-relationship error at the sign-in screen.

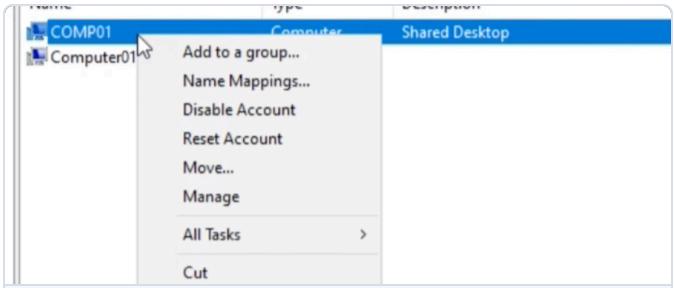


Figure 24. Removing the stale computer object in AD before rejoining the domain.

10. Findings & Recommendations

Table 3. Summary of findings and recommended actions.

#	Finding	Impact	Recommendation
1	Brute force over RDP succeeded against a weak password.	High — direct account compromise.	Enforce strong password & lockout policy; restrict/limit RDP exposure; add MFA where possible.
2	Local-account creation (T1136.001) generated no Splunk telemetry.	Medium — undetected persistence.	Enable and forward the relevant audit/Event ID (e.g., 4720) and build a detection rule.
3	Forwarder initially ran with insufficient privileges.	Medium — missing logs.	Run the forwarder with an identity scoped to read required logs; document the change.
4	Brute-force pattern (20×4625 in one instant) is highly detectable.	Positive — strong signal.	Create a correlation alert on failed-logon bursts to flag attacks in real time.

Overall, the environment demonstrates a working detection pipeline with one clearly identified and prioritized gap. The recommendations above convert the validation results directly into a short, actionable backlog.

11. Tools & Technologies

OPERATING SYSTEMS & VIRTUALIZATION

Oracle VirtualBox · Windows Server 2022 · Windows 10 · Ubuntu Server (Linux) · Kali Linux

ACTIVE DIRECTORY & WINDOWS ADMINISTRATION

AD Domain Services · Domain Controller · DNS · Group Policy (GPO) · Organizational Units · Security Groups · NTFS permissions · ADUC · Event Viewer · Remote Desktop (RDP)

SIEM, LOGGING & DETECTION

Splunk Enterprise · Splunk Universal Forwarder · Sysmon · SPL · Windows Event Logs · `inputs.conf` · MITRE ATT&CK

NETWORKING & COMMAND LINE

NAT networking · static IP · Netplan · TCP/IP & subnetting · PowerShell · Linux CLI · `Get-FileHash` (SHA-256) · `ping` / `ipconfig` / `ip a`

OFFENSIVE TESTING & VALIDATION

Crowbar (RDP brute force) · password wordlists · Atomic Red Team · Microsoft Defender

PRACTICES & CONCEPTS

Network diagramming · least privilege / RBAC · account lifecycle · incident troubleshooting · documentation · file-integrity verification

12. Skills & Lessons Learned

- **How Active Directory really works** — promoting a controller, structuring an organization with OUs, and why group-based permissions scale where per-user permissions do not.
- **Build logging, then prove it works** — monitoring must be validated against real activity, not assumed; confirming events actually arrive is part of the job.
- **Reading Windows security events** — distinguishing 4625 from 4624 and recognizing an attack pattern from the logs alone.
- **Testing defenses reveals blind spots** — a technique that produces no telemetry is itself a finding and a roadmap for the next detection.
- **Security is about reducing standing access** — least privilege, logon hours, and automatic expiration lower risk without relying on anyone to remember cleanup.
- **Fix the source, not the symptom** — recurring lockouts and broken domain trust require root-cause diagnosis, not repeated quick fixes.

These hands-on skills map directly to the requirements of IT Support, Help Desk, Junior System Administrator, Active Directory Administrator, and Tier-1 SOC Analyst roles, and align with the CompTIA A+, Network+, Security+, and Microsoft certification paths.

13. Conclusion

This project delivered a complete Active Directory environment and demonstrated the full lifecycle around it: deliberate design, careful deployment, instrumented monitoring, honest validation, targeted hardening, and practical troubleshooting. The controlled attack was not an end in itself but a means to prove the detection pipeline works and to expose where it does not — turning a home lab into evidence of real operational capability.

The result is a single environment that shows the ability to stand up infrastructure, monitor it, think like an attacker, verify defenses, and resolve the everyday incidents that keep a domain running.

13. Appendix A — Command & Event Reference

Table 4. Key commands and events referenced in this report.

Item	Context	Purpose
`Get-FileHash`	PowerShell	Verify installer integrity (SHA-256)
`ip a / ipconfig`	Linux / Windows	Confirm addressing
`crowbar -b rdp …`	Kali	RDP brute-force test
`EventCode 4625`	Splunk / Windows	Failed logon (brute-force indicator)
`EventCode 4624`	Splunk / Windows	Successful logon
`T1136.001`	MITRE ATT&CK	Create local account (visibility-gap test)
`:9997`	Splunk	Forwarder receiving port